Youssof Ahmed Aly

Sheikh Zayed, Egypt | youssof.ahmed.aly@gmail.com
linkedin.com/in/youssof-ahmed-13bb33241 | github.com/Dolphiny-1412

Summary

Computer Science graduate holding a dual degree from the University of Northampton (UK) and the Arab Academy for Science, Technology and Maritime Transport (Egypt). International cybersecurity & AI intern at MAHE–ISAC Centre of Excellence, Manipal, India. Graduation project in PPO reinforcement learning for HPC scheduling. Placed in the top 20% at the DeepX AI Competition (30 of 150 teams). Immediately available for roles in cybersecurity, data science/ML, and software engineering.

Education

B.Sc. Computer Science — University of Northampton (UK) Oct 2022 – Jul 2026

Dual award in partnership with the Arab Academy for Science, Technology & Maritime Transport, Egypt

Relevant coursework: Cybersecurity, Deep Learning, NLP, Digital Image Processing, Computer System Security, Robotics Applications, Operating Systems, Data Structures & Algorithms, Computer Networks, Database Systems, Computer Architecture, Software Engineering, Theory of Computation, Artificial Intelligence, Computer Graphics

High School Diploma (IGCSE) — Ethos International School 2017 – 2022

Experience

Cybersecurity & AI Intern Jul 2025 – Sep 2025

MAHE–ISAC Centre of Excellence for Cybersecurity, Manipal Academy of Higher Education — Manipal, India (via AIESEC)

- Built a fingerprint identification system in Python using image processing to classify fingerprint pattern types.
- Developed a signature verification model using machine learning to distinguish genuine signatures from forgeries.
- Developed a secure file upload service with MIME validation, ClamAV antivirus integration, HMAC-signed downloads, and per-IP rate limiting.
- Worked under the mentorship of Dr. Srikanth Prabhu (SoCE); received an official completion certificate from MIT Manipal.

Teaching Assistant Sep 2022 – Jan 2023

Ethos International School

- Supported classroom management and provided academic guidance to students requiring additional support.

Projects

Selected academic, internship and personal builds across cybersecurity, machine learning and systems programming.

AI-Driven HPC Cluster Scheduler — *Graduation project, team of 5*

Python, FastAPI, Celery, Redis, React, TypeScript, Stable-Baselines3

Full-stack HPC job scheduling platform driven by a PPO reinforcement learning agent that learns placement policies to reduce wait time and improve node utilisation. Built end to end: RL gym environment, scheduler core, REST API, and a live React/TypeScript dashboard. Supervised by Dr. Mohamed Mostafa El Taweel, AAST.

Fingerprint Identification System GitHub ↗

Python, OpenCV, scikit-learn

Classifies fingerprint pattern types (arch, loop, whorl) using image processing and ML. Built during the MAHE–ISAC internship.

Secure File Upload Service

Python, FastAPI, Docker, ClamAV

Side-by-side educational demo of vulnerable vs. hardened HTTP file upload. The hardened service implements MIME sniffing, ClamAV scanning, HMAC-signed downloads, per-IP rate limiting, and randomised storage.

Cryptographic Protocol Analyzer — *University computer security project*

Python, Scapy, Streamlit, Pandas, Plotly

Rule-based analyzer for PCAP/PCAPNG captures. Parses IPv4 TCP/UDP traffic with Scapy and flags insecure protocols (Telnet, FTP, HTTP, POP3, IMAP, SNMP and others), deprecated SSLv3/TLS 1.0/1.1 handshakes, and plaintext credential keywords, rating each finding Critical to Low with a remediation note. Includes a Streamlit dashboard with Plotly charts, a command-line interface, and CSV report export.

Virtual File System GitHub ↗

Python, SSH

Full-featured file system with a CLI shell, GUI, and SSH remote access, supporting full file and directory operations.

SysMonitor GitHub ↗

Python, Bash, Docker, Chart.js

Real-time CPU/GPU/temperature monitor with a GUI, Markdown reports, and a live web dashboard. Containerised with Docker.

Evil Dungeons GitHub ↗

C++, GLSL, CMake

Dungeon-crawler game with custom GLSL shaders and low-level graphics programming.

Technical Skills

Languages: Python, C, C++, Bash, JavaScript, TypeScript, SQL
Cybersecurity: Wireshark, MIME validation, ClamAV, HMAC signing, secure file handling, MAHE–ISAC CoE training
AI / Data Science: scikit-learn, OpenCV, Stable-Baselines3 (PPO), sentiment analysis, deep learning, NLP, image processing
Systems & Tools: Linux, Docker, Git, FastAPI, Celery, Redis, React, Streamlit, SQLite, SSH, Chart.js
Embedded & Hardware: Digital logic design, computer architecture, robotics applications

Achievements & Certifications

- Cybersecurity & AI Internship Certificate — MAHE–ISAC Centre of Excellence for Cybersecurity, Manipal Institute of Technology (Sep 2025)
- DeepX AI Competition — Top 30 of 150 teams (top 20%), Nile University (2025)

Languages

Arabic (native) | English (fluent, IGCSE educated)

Portfolio: dolphiny-1412.github.io